

VIP C2 PROGRAMME — FULL STACK DEVELOPMENT WITH
MERN

Proposed Solution

Online Complaint Registration and Management Systems — MERN
Stack E-commerce Application

Phase: Project Design Phase

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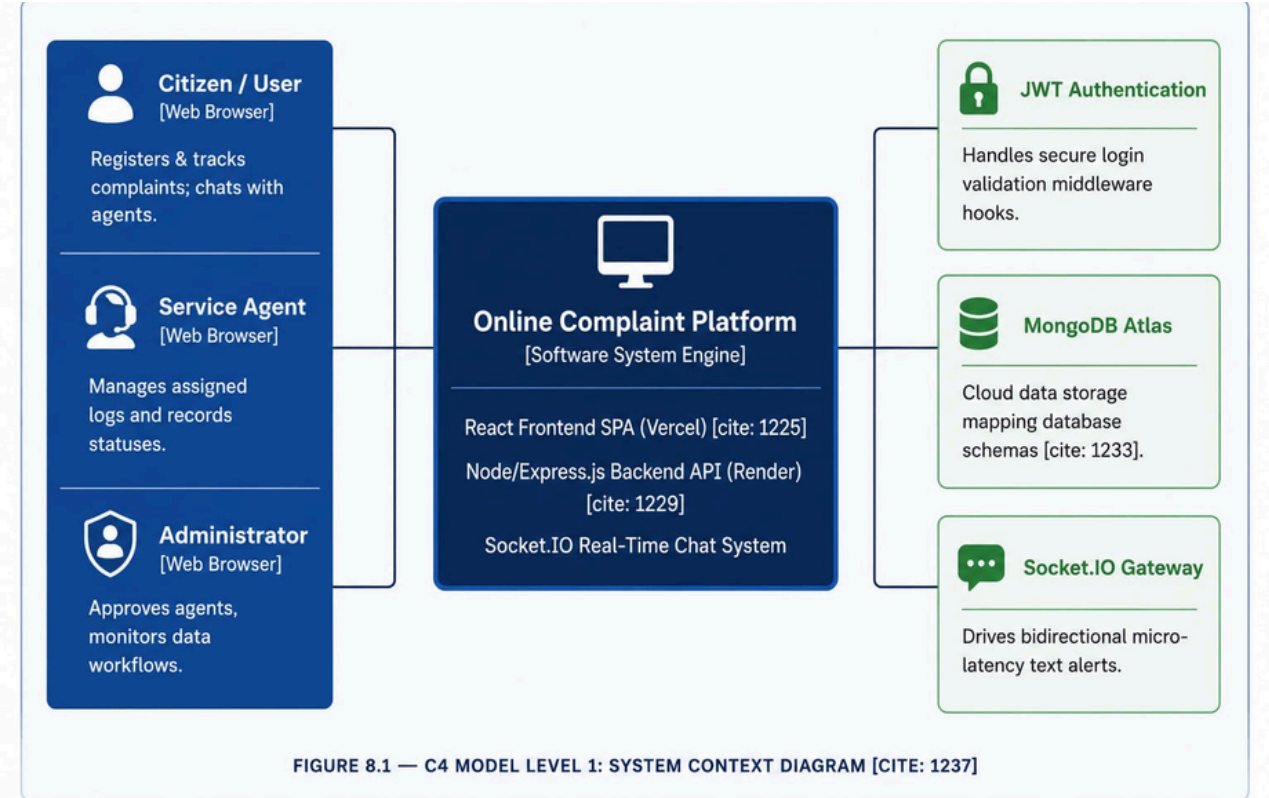
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Classification: Project Documentation — For Academic Submission

Proposed Solution

1. System Context Diagram



2. Solution Parameters

S.No.	Parameter	Detailed Description
1	Problem Statement	Citizens often face difficulties in registering complaints, tracking complaint progress, receiving timely updates, and communicating with responsible authorities. Traditional complaint handling systems suffer from poor transparency, delayed responses, fragmented communication channels, and inefficient complaint resolution workflows. Administrators face challenges in monitoring complaint activities, assigning complaints effectively, maintaining accountability, and generating meaningful performance insights for decision-making.
2	Solution Description	The Online Complaint Registration and Management System (OCRMS) is a full-stack MERN web application designed to streamline complaint management through a centralized digital platform. Citizens can register complaints, monitor complaint status, communicate with assigned agents through real-time chat, receive notifications, and provide feedback after resolution. Agents can manage assigned complaints and update complaint statuses. Administrators can manage users, approve agents, assign complaints, monitor complaint progress, and analyze system-wide performance through analytics dashboards
3	Novelty / Uniqueness	The solution combines complaint registration, complaint lifecycle management, real-time communication, notifications, analytics, and feedback collection within a single integrated platform. The system leverages Socket.IO for instant communication, JWT-based authentication for secure access control, and role-based authorization supporting Citizens, Agents, and Administrators. Cloud-based deployment ensures accessibility, scalability, reliability, and improved user experience across devices and locations
4	Social Impact / Customer Satisfaction	The platform improves transparency, accountability, and responsiveness in complaint handling. Citizens gain confidence through visibility into complaint progress and receive timely updates throughout the resolution process. Agents benefit from streamlined workflows and centralized information.

S.No.	Parameter	Detailed Description
		Overall, the system enhances trust, reduces complaint resolution time, improves service quality.
5	Business Model (Revenue Model)	The primary objective of the project is service improvement rather than revenue generation. However, future deployments may adopt a Software-as-a-Service (SaaS) model for municipalities, educational institutions, corporate grievance systems, housing communities, and public service organizations.

3. High-Level Feature Summary

Feature Category	Key Capabilities
Authentication & Security	JWT Authentication, Password Encryption, Secure Login, Role-Based Access Control (Citizen / Agent / Administrator), Session Management
Complaint Registration	Complaint Submission, Category Selection, Complaint Description, Evidence Upload, Complaint Tracking
Complaint lifecycle Management	Complaint Assignment, Status Updates, Resolution Workflow, Complaint Monitoring, Resolution History
Real-Time Communicationt	Socket.IO Real-Time Chat between Citizens and Agents, Instant Messaging, Query Resolution Support
Notification Systems	Real-Time Complaint Status Notifications, Assignment Notifications, Resolution Alerts, User Updates
Agent Operations	Assigned Complaint Management, Status Updates, Complaint Investigation, Citizen Communication
Administration Operations	User Management, Agent Approval Workflow, Complaint Assignment, Complaint Monitoring, Complaint Escalation
Analytics Dashboard	Complaint Statistics, Complaint Status Metrics, User Metrics, Agent Performance Monitoring, Resolution Reports

4 .Conclusion

The Online Complaint Registration and Management System (OCRMS) provides an efficient and user-friendly solution for modern complaint handling. By integrating complaint registration, tracking, real-time communication, notifications, and analytics into a single platform, the system improves transparency, accountability, and service quality. Its secure and scalable MERN-based architecture ensures reliable complaint management while supporting future enhancements and organizational growth